

Technology Stack (Architecture & Stack)

Date	27/09/2026
Team ID	SWTID-2026-8900
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Technical Architecture:

Architecture Flow Description:

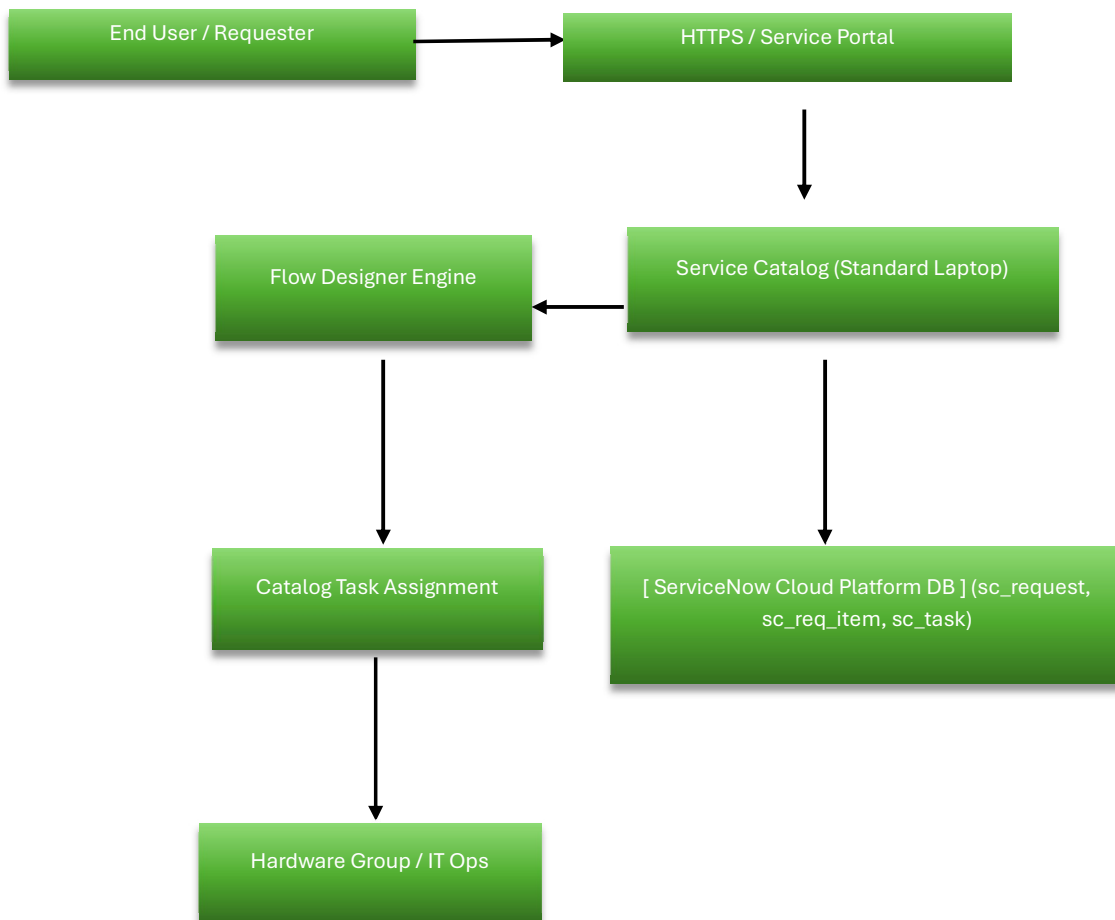


Table-1: Component & Technology Details

S.No	Component	Description	Technology
1.	User Interface	Self-Service Portal and Service Catalog interface for end users to request standard laptops and view order status.	ServiceNow Service Portal (AngularJS, HTML5, CSS3, Bootstrap Framework)
2.	Application Logic-1	Workflow Automation Engine that executes the "Standard Laptop Task" flow upon approval.	ServiceNow Flow Designer Engine (No-Code Flow Logic)
3.	Application Logic-2	Catalog Item configuration engine mapping process rules to the Standard Laptop item.	ServiceNow Process Engine / Service Catalog Engine
4.	Application Logic-3	Automated Catalog Task creation logic that sets Short Description and assigns tasks to the Hardware group.	ServiceNow Flow Designer Actions (Create Catalog Task Action)
5.	Database	Platform database storing Catalog Items, Service Requests, Requested Items (RITM), and Catalog Tasks	ServiceNow Relational Database (MariaDB / MySQL enterprise engine)
6.	Cloud Database	Enterprise Cloud Data Storage managed by ServiceNow PaaS infrastructure.	ServiceNow Cloud Infrastructure Storage
7.	File Storage	Storage for workflow logs, execution contexts, and attachments associated with Service Requests.	ServiceNow Attachment Storage / Sys_Attachment Table

8.	External API-1	Outbound notification service sending approval and assignment alerts to users and Hardware fulfillment teams.	ServiceNow REST / SOAP Email Integration (SMTP Service)
9.	External API-2	User Identity and Single Sign-On (SSO) integration for authentication.	SAML 2.0 / Azure Active Directory (Azure AD)
10.	Machine Learning Model	<i>Not Applicable</i> (Standard workflow automation solution using deterministic Flow Designer logic).	N/A
11.	Infrastructure (Server / Cloud)	Cloud-hosted Software-as-a-Service (SaaS) / Platform-as-a-Service (PaaS) deployment provided by ServiceNow.	ServiceNow Enterprise Cloud Infrastructure (Multi-Instance Architecture)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Front-end web application tools and libraries rendering the Service Catalog portal.	AngularJS, Bootstrap, JavaScript ES6
2.	Security Implementations	Access control lists (ACLs), role-based permissions (itil, catalog_admin), and encrypted data transport.	TLS 1.3 Encryption, SAML 2.0 SSO, ServiceNow ACLs & Role-Based Access Control
3.	Scalable Architecture	Multi-instance SaaS architecture separating workflow execution per customer instance with isolated database resources.	ServiceNow Multi-Instance Cloud Architecture
4.	Availability	Cloud deployment with automated failover, redundant data centers, and continuous platform monitoring.	High-Availability Dual Data Center Architecture (99.9% Uptime)
5.	Performance	Fast execution of Flow Designer triggers and task generation within milliseconds upon approval.	In-Memory Flow Processing, ServiceNow GlideRecord Query Optimization, CDN for static Portal Assets